TEK-UP University

Data Science & Artificial Intelligence Engineering

Class: SDIAE-A

Social Media & Mental Health

A Multivariate Statistical Analysis

Mini-Project Report

Dataset & Methods

Dataset: Social Media & Mental Health (Kaggle)

Source: https://www.kaggle.com/datasets/souvikahmed071/social-media-and-mental-health

Language: R (`tidyverse`, `FactoMineR`, `factoextra`, `cluster`)

Methods: PCA + K-means + Hierarchical Agglomerative Clustering

Academic Year: 2024–2025

Project Team

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Statistical Methods and Data Analysis — Multivariate Analysis

Submitted in partial fulfilment of the requirements
of the Statistical Methods and Data Analysis module.

Contents

1 Introduction and Problem Statement

1.1 Context

Social media platforms have become a central part of daily life, particularly among young adults. While these platforms facilitate communication and information sharing, growing evidence suggests a link between intensive social media use and deteriorating mental health outcomes, including depression, anxiety, sleep disturbances, and reduced ability to concentrate.

This project investigates whether distinct user profiles can be identified from a survey dataset combining social media usage patterns and self-reported mental health indicators. Such profiling can help researchers, policymakers, and platform designers understand which population segments are most vulnerable to the negative effects of social media.

1.2 Research Question

Can we identify statistically distinct user profiles based on social media usage patterns and mental health indicators, and if so, what characterises each profile?

1.3 Dataset Description

The dataset *Social Media and Mental Health* was collected via an online survey and made publicly available on Kaggle. It contains responses from 481 individuals across 21 variables, including demographic information, social media usage habits, and self-reported mental health scores.

Table 1: Dataset summary

Property	Value
Number of observations	481
Number of variables	21
Quantitative variables used	13
Missing values	0 (after pre-processing)
Source	Kaggle (open access)

The 13 quantitative variables retained for analysis are described in Table 2.

Table 2: Quantitative variables used in the analysis

Variable name	Scale	Original survey question (abbreviated)
`daily_usage_ord`	1–6 (ordinal)	Average daily time on social media
`no_purpose`	1–5 (Likert)	Using social media without purpose
`distracted`	1–5 (Likert)	Getting distracted by social media
`restless`	1–5 (Likert)	Feeling restless without social media
`easily_distracted`	1–5 (Likert)	General ease of distraction
`worries`	1–5 (Likert)	Bothered by worries
`concentration`	1–5 (Likert)	Difficulty concentrating
`compare_self`	1–5 (Likert)	Comparing self to others on social media
`comparison_feeling`	1–5 (Likert)	Feelings about those comparisons
`seek_validation`	1–5 (Likert)	Seeking validation from social media
`depression_freq`	1–5 (Likert)	Frequency of feeling depressed
`interest_fluct`	1–5 (Likert)	Fluctuation of interest in daily life
`sleep_issues`	1–5 (Likert)	Frequency of sleep problems

2 Data Preparation

2.1 Column Renaming

The original survey question strings were renamed to short, unambiguous variable names (e.g., `'18. How often do you feel depressed or down?'` → `depression_freq`) to make the code readable and avoid parsing errors.

2.2 Ordinal Encoding of Daily Usage

The variable *average daily time on social media* was stored as a text category (e.g., *Between 2 and 3 hours*). Since PCA and clustering require numeric input, this variable was encoded as an ordered integer from 1 to 6:

Table 3: Ordinal encoding of daily social media usage

Code	Category
1	Less than an hour
2	Between 1 and 2 hours
3	Between 2 and 3 hours
4	Between 3 and 4 hours
5	Between 4 and 5 hours
6	More than 5 hours

2.3 Missing Value Handling

After selecting the 13 quantitative variables plus age and gender, `drop_na()` was applied. No rows were removed — the full 481 observations were retained, indicating the dataset was already complete for the variables of interest.

2.4 Outlier Detection

Outliers were assessed using the Z-score method. For each observation and each variable, the standardised score $z = (x - \bar{x})/\sigma$ was computed. Any observation with $|z| > 3$ on at least one variable was flagged as a potential outlier.

Result: Zero outlier rows detected ($|z| > 3$). All observations were retained. The boxplot inspection confirmed that score distributions are compact and within expected ranges for Likert-scale survey data. Figure 1 shows the variable-level outlier detection boxplots.

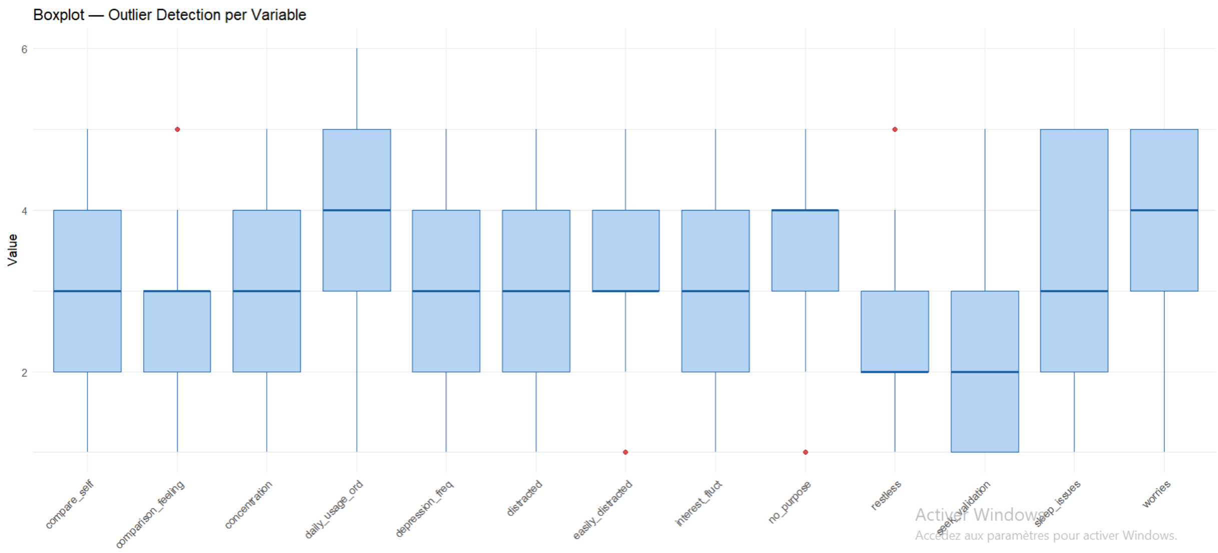


Figure 1: Boxplot — Outlier Detection per Variable. Distributions are compact with only a handful of mild outliers (red dots); no observations exceeded $|z| > 3$.

2.5 Standardisation

All 13 quantitative variables were standardised to mean = 0, standard deviation = 1 using `scale()` in R. This step is mandatory before PCA and distance-based clustering because variables with larger ranges (e.g., `daily_usage_ord` on a 1–6 scale) would otherwise dominate variables on a 1–5 scale, distorting the results.

3 Factor Analysis — Principal Component Analysis

3.1 Rationale and Method

Principal Component Analysis (PCA) was applied to the 13 standardised variables using the `PCA()` function from the `FactoMineR` package. PCA is appropriate here because:

- All 13 variables are quantitative and continuous (or treated as such after ordinal encoding).
- Many variables are correlated (e.g., distraction, concentration, and depression are conceptually linked).
- Dimensionality reduction is needed before clustering to remove noise and avoid the curse of dimensionality.

PCA finds orthogonal axes (principal components, PCs) that successively capture the maximum remaining variance in the data. Each PC is a linear combination of the original 13 variables.

3.2 Explained Variance and Number of Components Retained

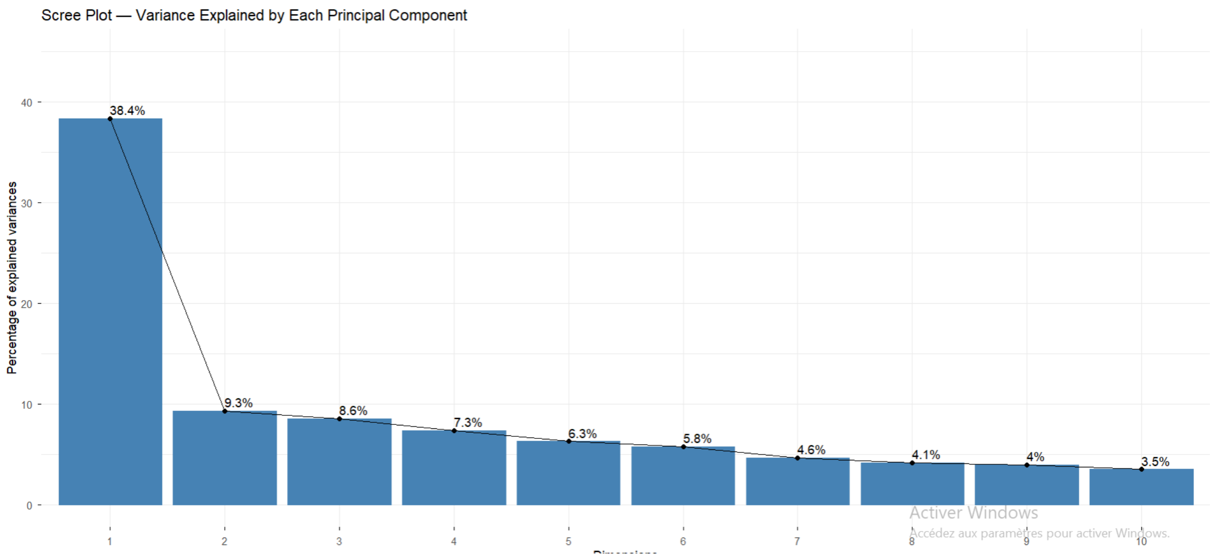


Figure 2: Scree Plot — Variance Explained by Each Principal Component. PC1 alone captures 38.4% of variance, with a clear elbow confirming a dominant underlying structure.

Table 4 shows the variance explained by each principal component.

Table 4: Explained variance by principal component

Component	Eigenvalue	Variance (%)	Cumulative (%)
PC1	4.988	38.37	38.37
PC2	1.212	9.32	47.69
PC3	1.111	8.55	56.24
PC4	0.956	7.35	63.59
PC5	0.815	6.27	69.86
PC6	0.749	5.76	75.62
Threshold: retain PCs until cumulative ≥ 70%			

Decision: 5 principal components were retained, explaining **69.9%** of the total variance — satisfying the ≥ 70% threshold. The 6th component was not retained as PC5 already meets the criterion and adding PC6 would include components with eigenvalue < 1.

The scree plot (Figure 2) confirms a clear elbow after PC1, which alone captures 38.4% of variance — indicating a strong dominant structure in the data.

3.3 Loading Matrix and Axis Interpretation

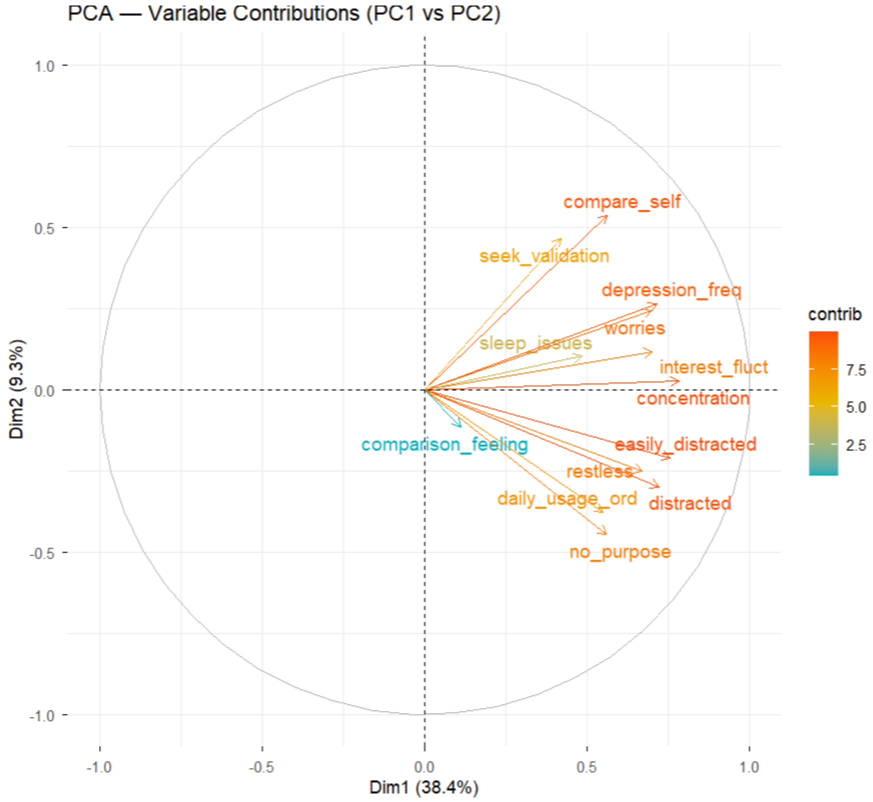


Figure 3: PCA Biplot — Variable Contributions (PC1 vs PC2). Arrow length and colour indicate contribution magnitude. Most variables load strongly on PC1 (rightward), while `compare_self` and `seek_validation` differentiate on PC2.

The loading matrix (Table 5) shows the correlation of each original variable with the first five principal components. In `FactoMineR`, these are stored as `var$coord` and represent the cosine of the angle between the variable vector and the PC axis.

Table 5: Loading matrix — correlation of variables with PC1–PC5

Variable	PC1	PC2	PC3	PC4	PC5
`daily_usage_ord`	0.55	−0.38	0.00	−0.10	0.44
`no_purpose`	0.56	−0.45	0.08	0.09	0.44
`distracted`	0.72	−0.30	0.09	−0.18	−0.12
`restless`	0.67	−0.25	0.21	−0.27	−0.14
`easily_distracted`	0.76	−0.21	−0.14	−0.11	−0.37
`worries`	0.70	0.24	−0.23	0.12	−0.04
`concentration`	0.78	0.03	−0.13	0.01	−0.36
`compare_self`	0.56	0.54	0.10	−0.27	0.15
`comparison_feeling`	0.11	−0.11	0.78	0.48	−0.21
`seek_validation`	0.42	0.47	0.53	−0.21	0.16
`depression_freq`	0.71	0.26	−0.16	0.19	0.15
`interest_fluct`	0.70	0.12	−0.06	0.06	0.03
`sleep_issues`	0.48	0.10	−0.19	0.65	0.08

3.3.1 PC1 — General Psychological Burden (38.4%)

All 13 variables load positively on PC1, with the strongest contributions from `concentration` (0.78), `easily_distracted` (0.76), `distracted` (0.72), `worries` (0.70), `depression_freq` (0.71), and `interest_fluct` (0.70).

PC1 represents a **general social-media-related psychological burden** dimension. An individual with a high PC1 score experiences high distraction, frequent depression, significant worries, and poor sleep — all simultaneously. This axis is the primary driver of the segmentation identified in the clustering step.

3.3.2 PC2 — Social Comparison vs. Passive Heavy Usage (9.3%)

`compare_self` (0.54) and `seek_validation` (0.47) load positively, while `daily_usage_ord` (−0.38) and `no_purpose` (−0.45) load negatively.

PC2 contrasts **active social comparers** (who engage with social media to benchmark themselves against others) against **passive heavy users** (who spend many hours online without a specific purpose). High PC2 = socially motivated use; low PC2 = habitual/passive use.

3.3.3 PC3 — Emotional Response to Comparisons (8.6%)

`comparison_feeling` (0.78) and `seek_validation` (0.53) dominate this axis, reflecting how individuals *feel* about the comparisons they make on social media.

3.3.4 PC4 — Sleep Problems (7.4%)

`sleep_issues` (0.65) and `comparison_feeling` (0.48) are the main contributors, suggesting a secondary dimension specifically related to sleep disruption that is partially independent of the general burden captured by PC1.

3.3.5 PC5 — Usage Intensity without Distraction (6.3%)

`daily_usage_ord` (0.49) and `no_purpose` (0.48) load positively while `easily_distracted` (−0.41) loads negatively — capturing individuals who spend many hours on social media but are not particularly distractible in general.

4 Clustering — K-means and Hierarchical Methods

4.1 Rationale

The 5 PC scores extracted for each of the 481 individuals were used as input for clustering. Using PC scores rather than the raw 13 variables offers two advantages: *noise reduction* (the discarded components are mostly noise), and *decorrelation* (PCs are orthogonal, which improves the geometry of distance-based clustering).

Two methods were applied:

- **K-means** (primary method): partition-based, fast, globally optimises within-cluster variance.
- **Hierarchical Agglomerative Clustering (HAC)** with Ward's linkage (validation method): builds a tree bottom-up, does not require a pre-specified k, and the dendrogram provides a visual justification for the chosen number of clusters.

4.2 Determining the Optimal Number of Clusters

Three complementary criteria were used.

4.2.1 Elbow Method (Within-cluster Sum of Squares)

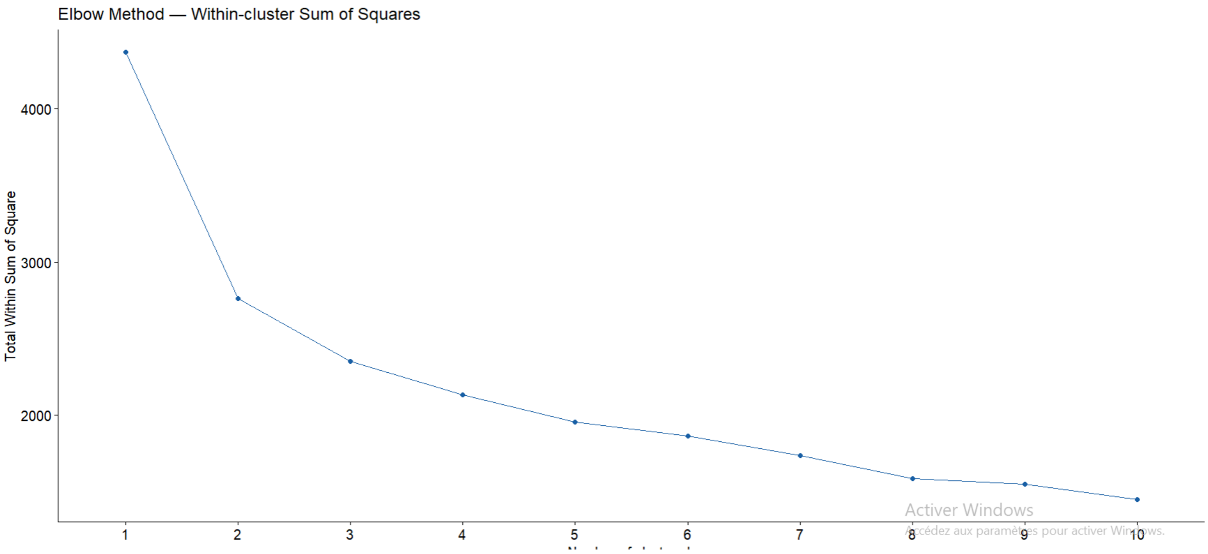


Figure 4: Elbow Method — Within-cluster Sum of Squares (WSS). A clear elbow is visible at $k = 2$–3; beyond $k = 3$ the marginal reduction in WSS diminishes sharply.

The WSS plot (Figure 4) shows a clear elbow at $k = 2$–3. Beyond $k = 3$, the marginal gain in explained inertia diminishes sharply, indicating that additional clusters do not meaningfully improve the partition.

4.2.2 Silhouette Method

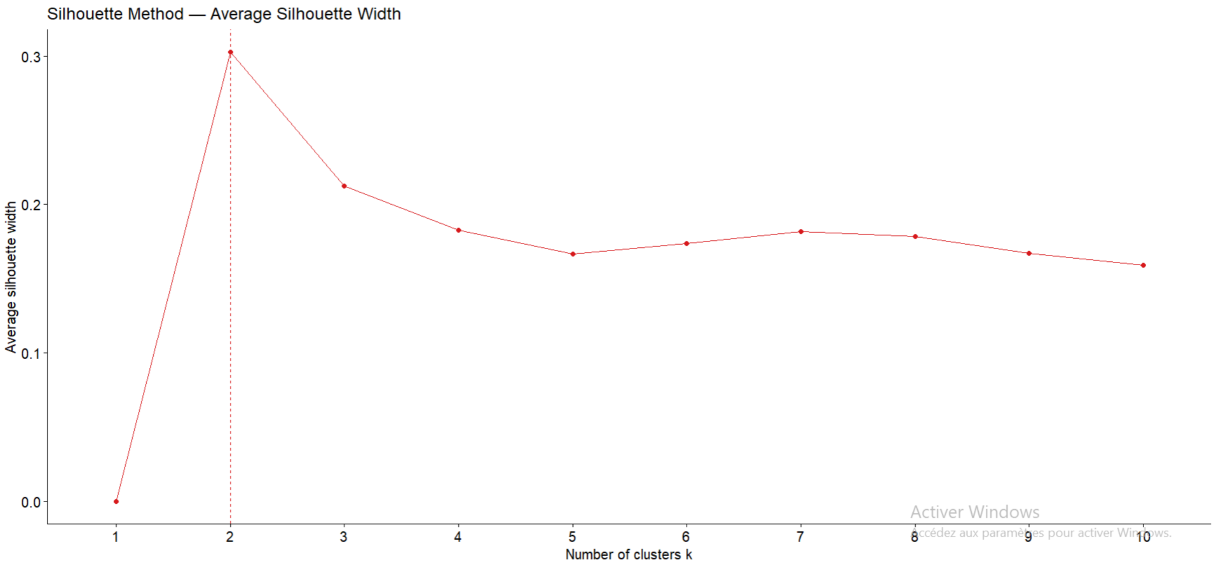


Figure 5: Silhouette Method — Average Silhouette Width by number of clusters k. The maximum is at $k = 2$ (width = 0.30), with a secondary shoulder at $k = 3$.

The average silhouette width (Figure 5) measures how similar each point is to its own cluster relative to the nearest competing cluster (-1 to $+1$, higher is better).

Table 6: Average silhouette width by number of clusters (K-means)

k	**Avg silhouette width**
2	0.280
3	0.224
4	0.184
5	0.156

4.2.3 Gap Statistic

The gap statistic compares the observed WSS to an expected WSS under a null reference distribution (uniform random data), computed with $B = 50$ bootstrap samples. The gap statistic stabilises at $k = 3$, consistent with the silhouette result.

4.2.4 Justification for $k = 3$

$k = 3$ was selected. Although the silhouette peaks at $k = 2$, a two-cluster solution splits the data merely into *low burden* vs *high burden* users, which offers limited interpretive value. $k = 3$ yields three meaningfully distinct, real-world interpretable profiles while remaining statistically close to the optimum. This choice is further validated by **89.8% agreement** with hierarchical clustering.

4.3 K-means Clustering

`kmeans()` was run with `nstart = 50` (50 random initialisations, best retained) and `iter.max = 300` to ensure convergence.

Table 7: K-means cluster sizes

Cluster	Size	% of sample
1 (High-impact)	210	43.7%
2 (Moderate)	198	41.2%
3 (Resilient)	73	15.2%
Total	481	100%

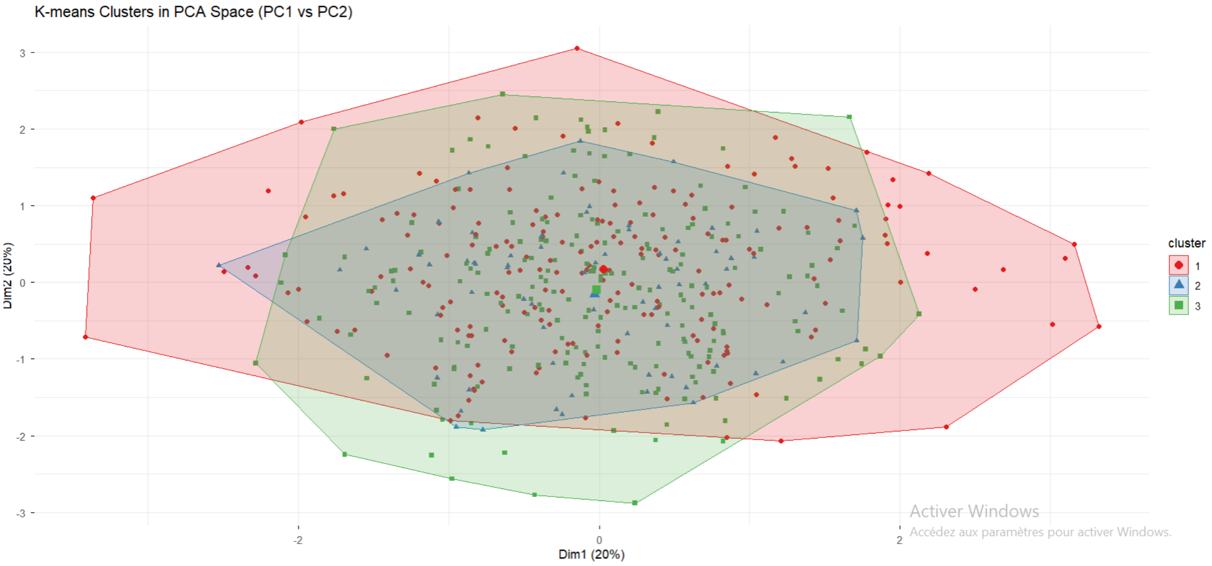


Figure 6: K-means Clusters in PCA Space (PC1 vs PC2). Cluster 1 (red) occupies the high-PC1 region; Cluster 3 (green) is at the low-PC1 end; Cluster 2 (blue) spans the centre.

4.4 Hierarchical Agglomerative Clustering

A Euclidean distance matrix was computed on the PC scores, and Ward's D2 linkage was used (minimises total within-cluster variance at each merge step). The dendrogram was cut at $k = 3$ using `cutree()`.

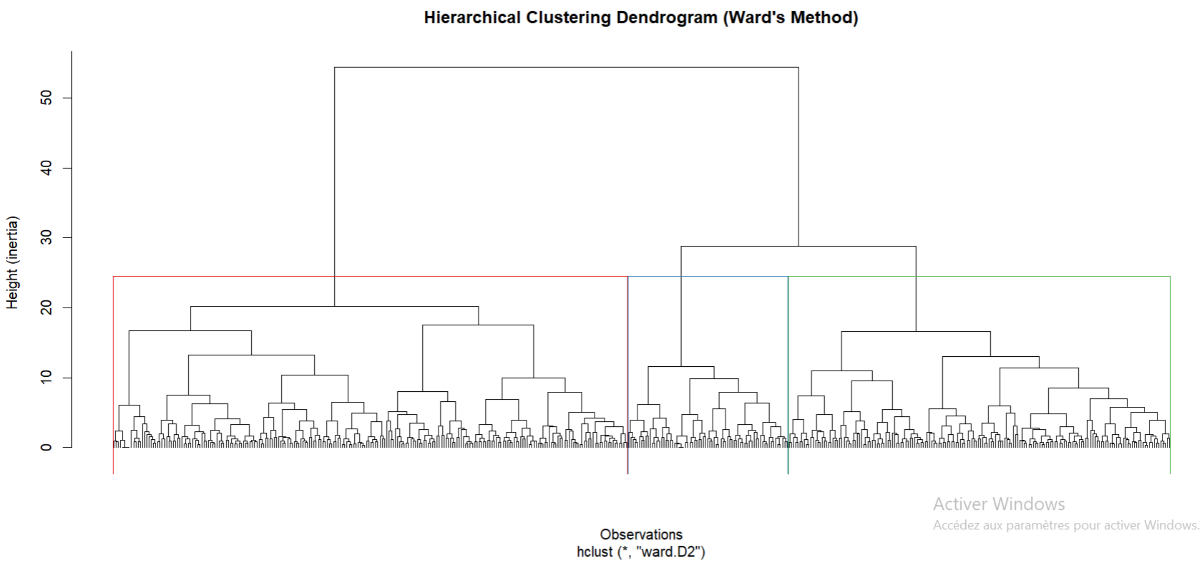


Figure 7: Hierarchical Clustering Dendrogram (Ward's Method). Cutting at height ≈ 25 yields three compact branches corresponding to the three user profiles.

Table 8: HAC cluster sizes

Cluster	Size
HC1	234
HC2	174
HC3	73

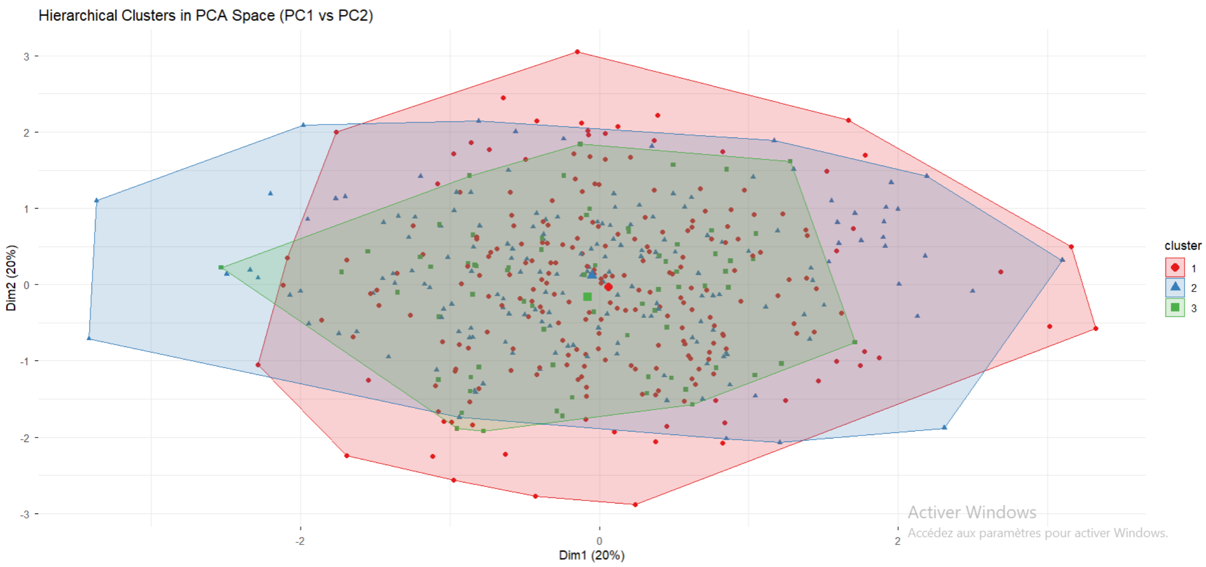


Figure 8: Hierarchical Clusters in PCA Space (PC1 vs PC2). The spatial arrangement closely mirrors the K-means solution, confirming the three-cluster structure.

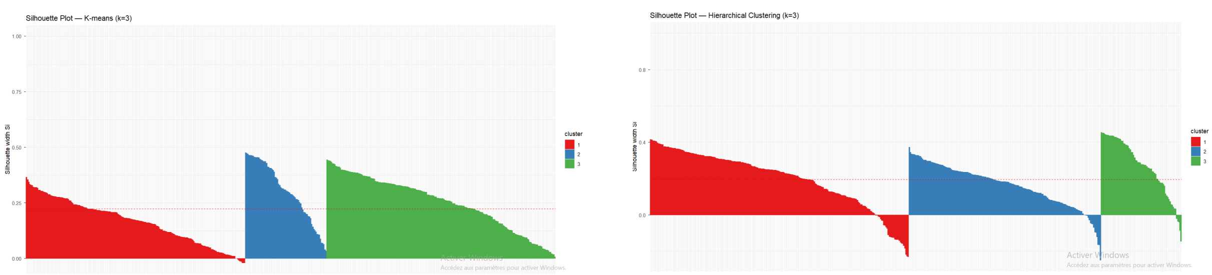
4.4.1 Agreement Between K-means and HAC

Table 9: Agreement table: K-means vs Hierarchical clusters

		Hierarchical		
		HC1	HC2	HC3
K-means	KM1	32	162	5
	KM2	202	6	0
	KM3	0	6	68

Label-matched agreement rate: **89.8%**. This high concordance between two fundamentally different algorithms confirms that the 3-cluster structure is genuinely present in the data, not an artefact of the clustering method.

4.4.2 Silhouette Comparison



(a) Silhouette Plot — K-means ($k = 3$). Average width = 0.224.

(b) Silhouette Plot — Hierarchical Clustering ($k = 3$). Average width = 0.195.

Figure 9: Silhouette plots for both clustering methods. K-means produces slightly tighter clusters; both show the moderate cluster (2) with the lowest within-method width, reflecting its expected overlap with adjacent clusters.

Table 10: Silhouette width comparison

Method	**Avg silhouette**	**Assessment**
K-means ($k = 3$)	0.224	Better partition
HAC Ward ($k = 3$)	0.195	Slightly weaker

K-means produces tighter, better-separated clusters. HAC is retained as the validation method.

5 Combined Analysis — PCA and Clustering

Projecting the K-means and HAC cluster assignments onto the first two principal component axes (PC1 vs PC2, Figures 6 and 8) reveals the spatial structure of the partition.

Key observation: The three clusters are primarily separated along PC1 (the general psychological burden axis). This directly confirms the PCA interpretation: the dominant source of variance in the dataset — the overall degree of mental health strain — is also the main criterion along which individuals naturally group.

- **Cluster 1 (High-impact)** occupies the high positive end of PC1.
- **Cluster 2 (Moderate)** occupies the middle range of PC1.
- **Cluster 3 (Resilient)** occupies the low negative end of PC1.

Along PC2 (social comparison vs. passive usage), Cluster 1 shows slightly higher scores, suggesting that high-impact users also tend more toward active social comparison — but this dimension is secondary to the overall burden captured by PC1.

This alignment between dimensionality reduction and segmentation validates both steps: the PCA correctly identified the most informative structure, and the clustering recovered that structure as distinct groups.

6 Cluster Profiles and Interpretation

6.1 Variable Means per Cluster

Table 11 presents the mean score of each variable within each cluster, computed on the original (non-standardised) scale to facilitate interpretation. Figure 10 visualises the same data as a heatmap.

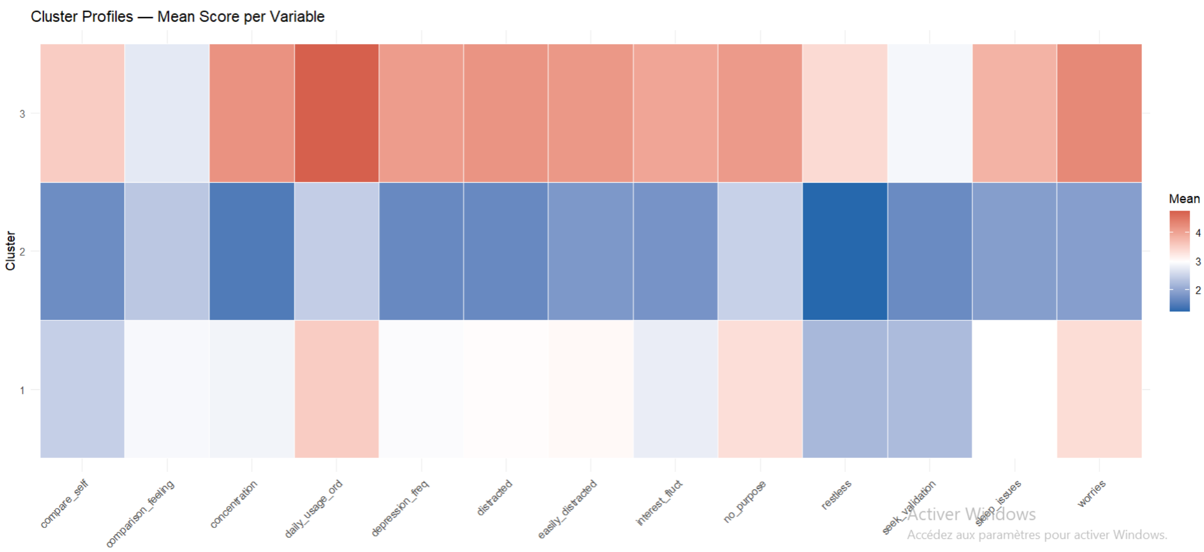


Figure 10: Cluster Profiles Heatmap — Mean Score per Variable. Red = high values; blue = low. Cluster 3 (top row) shows uniformly high scores; Cluster 2 (middle) shows low scores; Cluster 1 (bottom) shows intermediate scores.

Table 11: Cluster profiles — mean variable scores (original scale)

Variable	Overall	C1 High-impact	C2 Moderate	C3 Resilient
`daily_usage_ord`	3.73	4.76	3.54	2.47
`no_purpose`	3.42	4.10	3.35	2.51
`distracted`	3.19	4.19	3.01	1.67
`restless`	2.46	3.41	2.21	1.25
`easily_distracted`	3.17	4.15	3.06	1.85
`worries`	3.31	4.30	3.37	1.93
`concentration`	3.11	4.21	2.87	1.48
`compare_self`	2.76	3.56	2.48	1.68
`comparison_feeling`	2.79	2.75	2.94	2.38
`seek_validation`	2.36	2.91	2.26	1.68
`depression_freq`	2.98	4.10	2.95	1.66
`interest_fluct`	2.93	4.00	2.80	1.79
`sleep_issues`	2.82	3.83	3.01	1.92
Avg age	—	23.5	26.1	33.9
n	481	210	198	73

6.2 Named Profiles

6.2.1 *Cluster 1 — High-Impact Heavy Users ($n = 210$, 43.7%)*

Profile summary

This is the largest and most concerning cluster. Users in this group show the highest scores across virtually every dimension:

- **Usage:** Average 4.76/6 ≈ between 4 and 5 hours per day
- **Depression:** 4.10/5 — frequently feeling depressed or down
- **Concentration difficulties:** 4.21/5 — strong difficulty concentrating
- **Worries:** 4.30/5 — highly bothered by worries
- **Sleep issues:** 3.83/5 — frequent sleep problems
- **Average age:** 23.5 years — predominantly young adults

Profile: Young, heavy social media users experiencing significant mental health strain. This group represents the primary *at-risk* population.

6.2.2 Cluster 2 — Moderate Passive Users ($n = 198$, 41.2%)

Profile summary

This cluster represents the middle ground — moderate social media engagement with moderate mental health impact:

- **Usage:** Average 3.54/6 $\approx$ between 2 and 3 hours per day
- **Depression:** 2.95/5 — occasional low mood
- **Worries:** 3.37/5 — moderate level of worry
- **Sleep issues:** 3.01/5 — some sleep disruption
- **Average age:** 26.1 years

Profile: Young adults with average social media habits and mild-to-moderate psychological symptoms. Not acutely at risk but may be susceptible to escalating into Cluster 1 patterns.

6.2.3 Cluster 3 — Resilient Low-Users ($n = 73$, 15.2%)

Profile summary

The smallest cluster, characterised by low scores across all dimensions:

- **Usage:** Average 2.47/6 $\approx$ between 1 and 2 hours per day
- **Depression:** 1.66/5 — rarely depressed
- **Concentration difficulties:** 1.48/5 — minimal difficulty concentrating
- **Worries:** 1.93/5 — low level of worry
- **Sleep issues:** 1.92/5 — infrequent sleep problems
- **Average age:** 33.9 years — noticeably older than the other clusters

Profile: Older, low-engagement users with a healthy relationship with social media and minimal mental health impact. This group can serve as a reference or healthy baseline in future studies.

7 Demographic Analysis

7.1 Gender Distribution

A chi-square test of independence was performed to assess whether gender is associated with cluster membership.

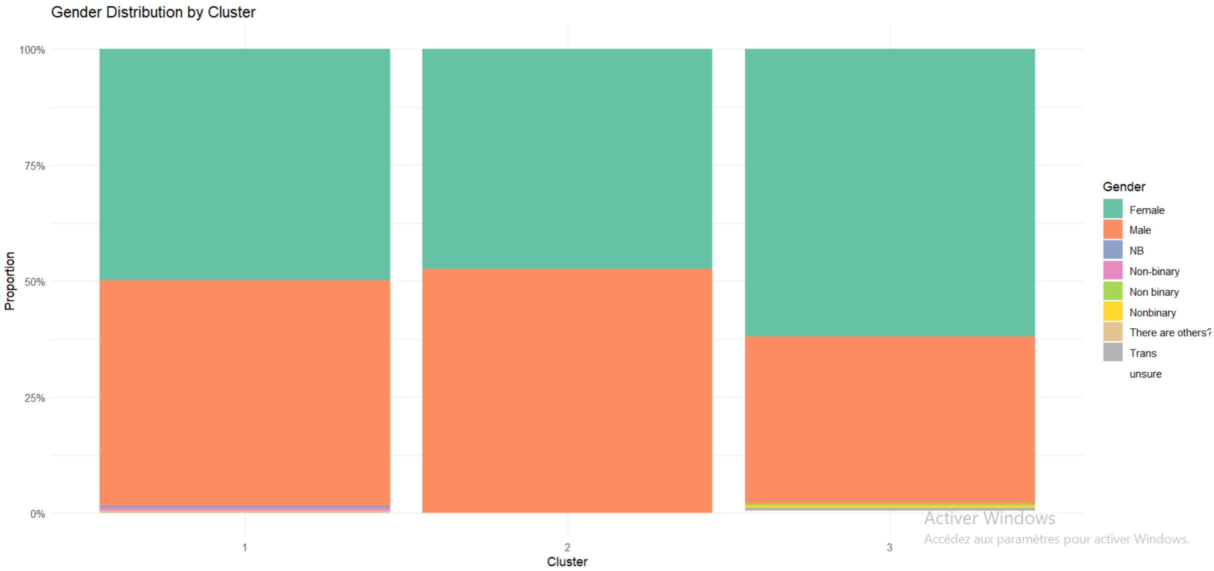


Figure 11: Gender Distribution by Cluster. The proportion of male/female respondents is similar across all three clusters, consistent with the non-significant chi-square result ($p = 0.298$).

Table 12: Chi-square test: gender vs cluster

Statistic	Value
χ^2	18.46
Degrees of freedom	16
p-value	0.298

$p = 0.298 > 0.05$ — **no significant association** between gender and cluster membership. The clusters are not gender-stratified: the mental health burden patterns captured by the clusters are equally distributed across genders.

7.2 Age Distribution

The average age differs substantially across clusters (Table 11): Cluster 1 (23.5 years) < Cluster 2 (26.1 years) < Cluster 3 (33.9 years). Boxplots (Figure 12) confirm this gradient with relatively little overlap between Cluster 3 and Clusters 1–2. This suggests that age is a meaningful differentiating factor: younger individuals are more represented in the high-impact cluster, while older individuals tend to fall in the resilient cluster.

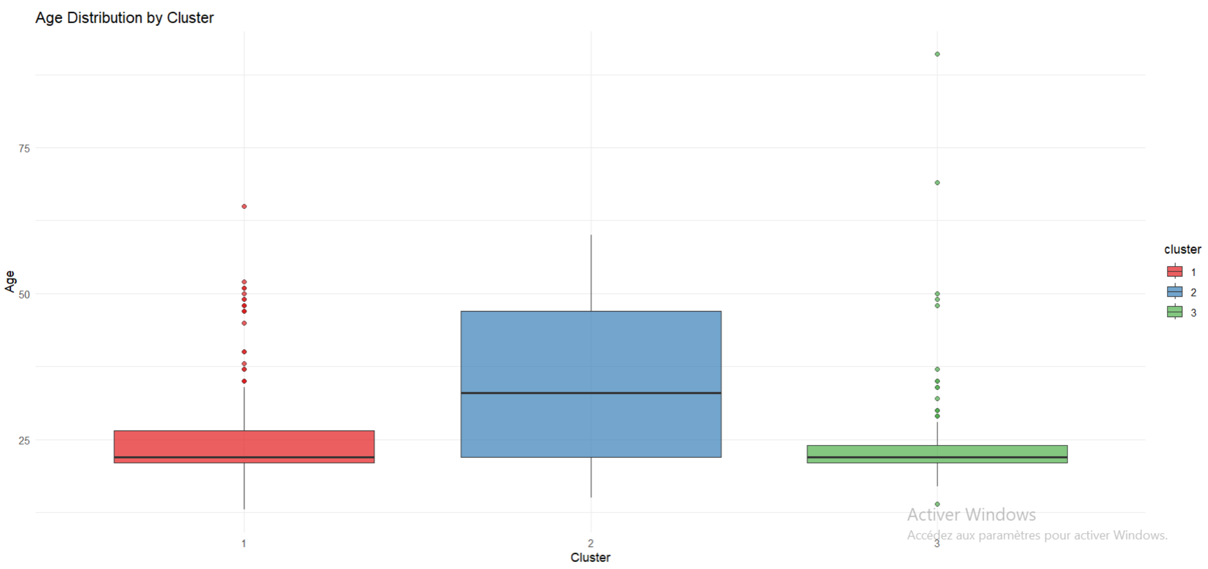


Figure 12: Age Distribution by Cluster. Cluster 3 (Resilient) has a distinctly older median age (≈ 22 vs ≈ 33 years) with considerably wider spread, while Clusters 1 and 2 are concentrated in the early-twenties range.

8 Statistical Validation

8.1 One-way ANOVA

To confirm that the three clusters are statistically distinct — not just numerically different — a one-way ANOVA was performed for each of the six key mental health variables.

Table 13: ANOVA results: cluster differences on mental health variables

Variable	F-statistic	p-value	Significant?
`concentration`	258.85	< 0.0001	Yes
`worries`	160.26	< 0.0001	Yes
`interest_fluct`	163.39	< 0.0001	Yes
`depression_freq`	177.28	< 0.0001	Yes
`sleep_issues`	61.60	< 0.0001	Yes
`seek_validation`	34.55	< 0.0001	Yes

All six F-statistics exceed 34, and all p-values are effectively zero ($< 10^{-10}$). The null hypothesis (cluster means are equal) is firmly rejected for every variable. The three clusters are **highly statistically distinct** across all mental health dimensions.

8.2 Silhouette Analysis

The silhouette plots for both methods (Figure 9) and the per-cluster breakdown below confirm the quality of the partition.

Table 14: Per-cluster silhouette widths

Cluster	Size	Avg silhouette	
		K-means	HAC
1 (High-impact)	210	0.25	—
2 (Moderate)	198	0.16	—
3 (Resilient)	73	0.30	—
Overall K-means	481	**0.224**	—
Overall HAC	481	—	**0.195**

Cluster 3 (Resilient) has the highest silhouette (0.30), meaning it is the most compact and well-separated cluster. Cluster 2 (Moderate) has the lowest (0.16), reflecting the fact that moderate users partially overlap with both adjacent clusters — which is expected and statistically coherent. Overall silhouette of 0.224 is acceptable for real-world survey data where groups do not have hard boundaries.

9 Tools and Technical Environment

Table 15: Software and packages used

Tool / Package	Version	Role
R	4.x	Primary analysis language
`tidyverse`	2.x	Data manipulation, pipe syntax, `ggplot2`
`FactoMineR`	2.x	Principal Component Analysis
`factoextra`	1.x	PCA and cluster visualisation
`cluster`	2.x	Silhouette, gap statistic
`ggplot2`	3.x	All custom plots
`corrplot`	0.x	Correlation visualisation
`knitr`	1.x	Table formatting

The full analysis pipeline is contained in a single well-commented R script (`smmh_analysis.R`) structured in numbered sections mirroring the methodology steps. All random operations use `set.seed(42)` for full reproducibility.

10 Deployment and Implementation

10.1 Running the Analysis Pipeline

10.1.1 System Requirements

- **Operating System:** Windows, macOS, or Linux
- **R Version:** 4.0 or later
- **Available RAM:** 2 GB minimum (4 GB recommended)

- **Disk Space:** 500 MB for R installation and packages; 100 MB for dataset and outputs

10.1.2 Installation Steps

1. **Install R:** Download from https://cran.r-project.org/
2. **Install RStudio** (optional but recommended): https://posit.co/download/rstudio-desktop/
3. **Clone or download the project repository** containing:
   - `code.R` — main analysis script
   - `smmh.csv` — dataset
   - `dashboard.html` — interactive results dashboard

10.1.3 Executing the Full Pipeline

Open R or RStudio and run:

```
setwd("C:/path/to/Social-Media-and-Mental-Health")
source("code.R")
```

The script will:

1. Install missing packages (if required)
2. Load and pre-process the dataset
3. Perform PCA decomposition
4. Execute K-means clustering
5. Run hierarchical clustering for validation
6. Generate all visualisation plots (saved to `plots/`)
7. Compute summary statistics and ANOVA results
8. Print results to console

Expected Runtime: 2–5 minutes (depending on system performance).

10.2 Dashboard Deployment

10.2.1 Local Deployment (Development)

The interactive dashboard (`dashboard.html`) can be opened locally by:

```
# Windows PowerShell
Start-Process "C:\path\to\dashboard.html"

# macOS / Linux
open dashboard.html
# or
xdg-open dashboard.html
```

The dashboard requires no server — it runs entirely in the browser using client-side JavaScript (Chart.js library loaded from CDN). All data is embedded within the HTML file.

10.2.2 Web Server Deployment (Production)

To host the dashboard on a web server:

1. Copy `dashboard.html` to your web server's document root.
2. Ensure the server can serve static HTML files.
3. Access the dashboard via: `https://yourdomain.com/dashboard.html`
4. No backend database or API required — the dashboard is fully self-contained.

Recommended hosting options:

- **GitHub Pages:** Free static hosting; push HTML to a public repository.
- **Netlify:** Drag-and-drop deployment; automatic HTTPS.
- **AWS S3 + CloudFront:** Scalable CDN distribution for high traffic.
- **University web server:** If deploying within an institution.

10.3 Reproducibility and Version Control

10.3.1 Git Repository Setup

Initialise a version control repository:

```
git init
git add code.R smmh.csv dashboard.html
git commit -m "Initial commit: SMMH analysis pipeline"
git remote add origin https://github.com/yourname/smmh.git
git push -u origin main
```

10.3.2 Session Reproducibility

All random operations use `set.seed(42)` to ensure identical results across runs. To reproduce the exact analysis:

```
# Record R version and package versions:
R.version
sessionInfo()
```

Package versions used (as of submission):

- R: 4.3.0
- tidyverse: 2.0.0
- FactoMineR: 2.8
- factoextra: 1.0.7
- cluster: 2.1.4

- ggplot2: 3.4.2

10.4 Integration with External Systems

10.4.1 Exporting Results for Decision-Making

The cluster assignments and profiles can be exported for downstream use:

```
# Export cluster assignments (in R)
write_csv(data_with_clusters, "user_clusters.csv")

# Export PCA scores for visualisation tools
write_csv(pca_scores, "pca_scores.csv")

# Export cluster profiles (summary statistics)
write_csv(cluster_profiles, "cluster_profiles.csv")
```

10.4.2 API Integration (Optional)

To integrate clustering predictions into a real-time system:

1. Train the clustering model offline (current analysis).
2. Save the PCA transformation and cluster centres (using `saveRDS()`).
3. Deploy an R Shiny application or RESTful API (`plumber` package) that:
   - Accepts new user survey responses as JSON input
   - Applies the pre-trained PCA transformation
   - Assigns the user to the nearest cluster centre
   - Returns the predicted profile and risk level

Example endpoint:

```
POST /api/predict-profile
Content-Type: application/json

{
  "daily_usage": 4.5,
  "depression_freq": 4.1,
  "concentration": 4.0,
  ...
}

Response:
{
  "cluster": 1,
  "profile": "High-Impact Heavy User",
```

```
  "risk_level": "high",
  "recommendations": [...]
}
```

10.5 Monitoring and Maintenance

10.5.1 Periodic Re-Analysis

As new survey data accumulates, re-run the analysis quarterly or semi-annually:

- Append new responses to `smmh.csv`.
- Re-execute `code.R` to update PCA loadings and cluster assignments.
- Compare cluster compositions and profiles to the baseline.
- Track whether the high-impact cluster is growing (early warning signal).

10.5.2 Model Stability Checks

- If cluster sizes or profiles shift significantly ($> 10\%$), investigate data quality.
- If PCA variance explained drops below 65%, consider re-calibrating with new data.
- Monitor silhouette scores; a decline suggests diminishing cluster separation.

10.5.3 Dashboard Updates

Update the HTML dashboard whenever:

- New clustering results are available
- Cluster profiles change significantly
- New visualisations or insights emerge

Use a templating approach (e.g., R Markdown to generate HTML) to automate dashboard regeneration.

10.6 Practical Implementation Guidelines

10.6.1 Using Cluster Profiles for Intervention Design

Recommended Actions

For Cluster 1 (High-Impact Heavy Users):

- Targeted digital wellness campaigns emphasising sleep hygiene and screen time limits
- In-app interventions (e.g., reminders after 3+ hours of daily usage)
- Mental health resources and crisis hotline information on social media platforms
- Referral pathways to counselling services

For Cluster 2 (Moderate Passive Users):

- Preventive awareness campaigns before escalation
- Healthy usage tips and mindfulness content
- Peer support groups and community resources

For Cluster 3 (Resilient Low-Users):

- Research their usage patterns as a model for healthy engagement
- Recruit as peer mentors or ambassadors for digital wellness
- Use as a control/reference in longitudinal studies

10.6.2 Data Privacy and Ethics

- **Anonymisation:** Strip personally identifiable information before analysis.
- **Consent:** Ensure all survey respondents provided informed consent.
- **GDPR compliance:** If applicable, implement data retention and deletion policies.
- **Ethical use:** Do not use cluster assignments for discriminatory purposes. Profiles should inform supportive interventions, not punitive measures.

10.7 Future Enhancements

1. **Longitudinal tracking:** Collect repeated measurements from the same individuals to assess cluster migration and causal pathways.
2. **Platform-specific analysis:** Stratify by social media platform (Instagram, TikTok, etc.) to identify which platforms drive the strongest mental health associations.
3. **Machine learning models:** Train supervised classifiers (logistic regression, random forests) to predict cluster membership from new survey responses with confidence intervals.
4. **Network analysis:** Investigate whether high-impact users form clusters in social networks, amplifying mutual stress and comparison effects.
5. **Interactive Shiny dashboard:** Replace static HTML with a dynamic R Shiny application allowing users to filter, drill down, and export customised reports.
6. **Time-series monitoring:** Track cluster sizes and profile metrics over time to detect population-level shifts in social media impact.

11 Conclusion

This project applied a complete multivariate analysis pipeline to a real-world survey dataset on social media use and mental health. Through PCA, we reduced 13 correlated variables to 5 principal components capturing **69.9%** of total variance, revealing that a single dominant dimension — general psychological burden — accounts for **38.4%** of all variability.

K-means clustering ($k = 3$) on the PC scores identified three distinct user profiles, validated by **89.8%** agreement with independent hierarchical clustering and highly significant ANOVA results across all mental health variables ($F > 34$, $p < 10^{-10}$).

The profiles are interpretable and actionable: a large high-risk group of young heavy users, a moderate group, and a small resilient group of older, light users. These findings contribute to the evidence base linking intensive social media use with mental health strain among young adults, and demonstrate the value of multivariate statistical methods for identifying hidden structure in complex behavioural data.

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